Mukesh Patel School of Technology Management & Engineering
School of Technology Management and Engineering

Course Policy Document

Course Name - Digital Logic Design **(**702EX0C028**)**

<table>
<tr><td>Program and Semester:
B Tech IT
MBA Tech IT</td><td>Pre-requisite Course :
Knowledge of Basic Electronics</td></tr>
<tr><td>Academic Year: 2025-26</td><td>Credit Details:
<table>
<tr><td>L</td><td>T</td><td>P</td><td>C</td><td>H</td></tr>
<tr><td>2</td><td>0</td><td>2</td><td>3</td><td>6</td></tr>
</table></td></tr>
<tr><td>Name of Course Faculty: Prof. Ketki Deshmukh
Program: Electronics and Telecommunication
Contact Details: 022-4502-4773
Office Hours: 10AM to 5PM</td><td>Faculty associated with the course: -
1. Dr Atul Patil (Shirpur Campus)
2. Prof Ketki Deshmukh (Mumbai Campus)</td></tr>
<tr><td colspan="2">Pre-Course Activity:
Course policy discussion
Discuss significance and multidisciplinary application of the course,
Provide with reading material links related to unit 1to enable students to come prepared to class.</td></tr>
<tr><td colspan="2">Course link: To be provided</td></tr>
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1. Introduction to the Course

1.1 Importance of the Course

1.1.1 Domain Relevance:

Digital Logic Design is essential in engineering, particularly in computer, electrical, and electronics engineering. It underpins the creation and analysis of digital systems, playing a key role in computer architecture (CPUs and memory units) and embedded systems (microcontrollers in smartphones and automotive devices). The principles are crucial for circuit design, optimization, control systems, and communication systems, as well as FPGA and VLSI design for custom hardware.

1.1.2 Industry Relevance:

Digital Logic Design is highly relevant in industries such as technology, automotive, telecommunications, and consumer electronics. It underpins the creation and optimization of digital systems like CPUs, microcontrollers, and embedded systems found in smartphones, cars, and home appliances. The principles of digital logic are crucial for designing circuits, control systems, communication devices, and custom hardware solutions like FPGAs and VLSI. Engineers with expertise in digital logic are in high demand for developing reliable, efficient, and innovative digital products, making this subject essential for driving technological advancements across various industries.

1.2 Objectives of the Course:

1.2.1 Foundation in Digital System: Equip students with a thorough understanding of the fundamental principles of digital systems, including Boolean algebra, logic gates, and combinational and sequential circuits. This foundational knowledge is crucial for designing and analyzing complex digital systems and devices.

1.2.2 Practical Circuit Design Skills: Develop practical skills in designing, optimizing, and implementing digital circuits. This includes hands-on experience with tools and techniques used in the creation of digital components such as multiplexers, decoders, flip-flops, counters, and memory units.

1.2.3 Problem Solving and Analytical Skills: Enhance students' problem-solving and analytical abilities by teaching them how to systematically approach and solve complex digital design problems, which includes logical thinking, troubleshooting, and the ability to design efficient, reliable, and optimized digital solutions for real-world applications.

2. Course Outcomes (CO), Mapping with Program Outcomes (PO), and Program Specific Outcomes (PSO)

2.1 Course Outcomes

2.1.1 CO1: Translate different number systems and binary codes, including weighted and non-weighted codes. *(L2 – Understanding)*

2.1.2 CO2: Simplify Boolean expressions using De Morgan's Theorem, SOP, POS, and Karnaugh maps. *(L4 – Analyzing)*

2.1.3 CO3: Build combinational logic circuits such as adders, multiplexers, decoders, and encoders. *(L3 – Applying)*.

2.1.4 CO4: Analyze sequential circuits, including flip-flops, registers, and counters. *(L4 – Analyzing)*

2.1.5 CO5: Make use of VHDL for realizing basic digital logic circuits in different modeling styles. *(L3 – Applying)*

2.2 Program Outcomes (PO) the course contributed to:

2.2.1 PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

2.2.2 PO2: Problem Analysis: Identify, formulate, review research literature and analyse complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

2.2.3 PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

2.2.4 PO-4: Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

2.2.5 PO-5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

2.2.6 PO-6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its

impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

2.2.7 **PO-7: Ethics**: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

2.2.8 **PO-8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

2.2.9 **PO-9: Communication**: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

2.2.10 **PO-10: Project management and finance**: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

2.3 **PO-11: Life-long learning:** Recognize the need for, and have the preparation and ability for independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

2.4 Program Specific Outcomes (PSO):

2.4.1 PSO-1: **Apply elemental principles** to conceptualize, design, and implement innovative solutions for communication, networking, and signal processing applications

2.4.2 PSO-2: **Exhibit expertise and adaptability** in developing efficient and pioneering technologies in Embedded Systems, Artificial Intelligence, and Next-Generation Communication systems..

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2									2	3	
CO2	3	3	-	3							2	3	

CO3	3	2	3	-	3						2	3	3
CO4	3	3	-	3							2	3	2
CO5	3	2	3	-	3						3	3	3

Mapping Levels: 1- High, 2-Medium, 3-Low

2.5 Student Outcomes (SO) (For ABET accredited Programs):

2.5.1 **SO-1:** Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

2.5.2 **SO-2:** An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

2.5.3 **SO-3:** An ability to communicate effectively with a range of audiences

2.5.4 **SO-4:** An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal context.

2.5.5 **SO-5:** An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives

2.5.6 **SO-6:** An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions

2.5.7 **SO-7:** An ability to acquire and apply new knowledge as needed, using appropriate learning strategies

3. Teaching-learning methodology

3.1 Instruction Plan

Lecture No.	Topic	Teaching Method *	Blooms Level	Resources	Cos mapped	Assessment and Evaluation
Unit 1: Introduction to Digital Systems						
1	Introduction to Digital Systems and logic simplification Number Systems: binary, octal, hexadecimal, BCD	PPT, Whiteboard	UNDERSTAND	**TB1, TB2** TB1: Sec 1.1-1.5 pp 1-10 TB2: Sec 2.1-2.4 pp 28-33	CO1	Assignment, Class Test-1 and TEE

2	Different types of Conversion one number system to another Decimal to binary, Binary to decimal Decimal to octal, Octal to decimal Decimal to Hex, Hex to decimal Hex to binary, binary to hex Octal to binary, Binary to octal Octal to Hex and Hex to octal.		UNDERSTAND	**TB1, TB2,** RB1 TB1: Sec 1.3-1.4 Pp 6-8 TB2: Sec-2.1-2.4 Pp 28-32, Sec 2.7-2.10 Pp 43-56 RB1: Sec-2.4 Pp 41-49	CO1	Assignment, Class Test-1 and TEE
3	Weighted codes: BCD and binary, non-weighted codes: grey and excess 3. Conversion from one code to another. Gray to binary code conversion and binary to grey code conversion. Binary to access 3 code conversion.		INTERPRET	TB2, RB1 TB2: Sec-2.10 Pp 57-63 RB1: Sec-3.1 Pp 62	CO1	Lab Submission Class Test-1 and TEE
4	Binary Addition and Binary Subtraction using 1's and 2's Complement method		INTERPRET	TB2 TB2: Sec-2.4-2.6 Pp 34-42	CO1	Lab Submission and Class Test-1 and TEE
Unit 2: Logic Gates and Boolean Algebra						
5	Logic gates like NOT, AND, OR, NAND, NOR, XOR and XNOR and implementation of these logic gates using universal gates	PPT, Whiteboard	ANALYZE, IMPLEMENT	TB1, TB2 TB1: Sec-2.8 Pp 60-65 TB2: Sec-1.3-1.5 Pp 3-14	CO1	Lab Submission and Class Test-1 and TEE
6	Review of Boolean Algebra and De Morgan's Theorem			TB1, TB2 TB1: Sec-2.3-2.4 Pp 40-50 TB2: Sec-1.6 Pp -15-17	CO2	
7	Review of Boolean Algebra and De Morgan's Theorem		APPLY	TB1, TB2 TB1: Sec-2.3-2.4 Pp 40-50 TB2: Sec-1.6 Pp -15-17	CO2	

8	SOP & POS forms, Canonical forms		APPLY	TB1, TB2, RB1 TB1: Sec-2.5-2.6, 3.4 Pp 46-57, 84 TB2: Sec-5.2 Pp -166-172 RB1: Sec-6.5 Pp 162-166	CO2	
9	Karnaugh maps up to 4 variables		SIMPLIFY	TB1, TB2, RB1 TB1: Sec-3.1-3.3 Pp 73-83 TB2: Sec-5.3- 5.8 Pp -173-190 RB1: Sec-6.7 Pp 171	CO2	
10	Karnaugh maps up to 4 variables		SIMPLIFY	TB1, TB2, RB1 TB1: Sec-3.1-3.3 Pp 73-83 TB2: Sec-5.3- 5.8 Pp -173-190 RB1: Sec-6.7 Pp 171	CO2	
11	**Mid Term Test I**					
12	Half and Full Adder-Truth table, Design using Karnaugh map and circuit diagram		DESIGN	TB1, TB2, RB1 TB1: Sec-4.1-4.6 Pp 125-144 TB2: Sec-5.6, 5.7,6.4,6.5 Pp -242-246 RB1: Sec-7 Pp 233-234	CO3	Lab Submission and Class Test-1 and TEE
Unit 3: Combinational logic circuit and its implementation						
13	Half and Full Subtractor-Truth table, Design using Karnaugh map and circuit diagram	PPT, Whiteboard	DESIGN	TB1, TB2, RB1 TB1: Sec-4.1-4.6 Pp 125-144 TB2: Sec-5.6, 5.7,6.4,6.5 Pp -242-246 RB1: Sec-7 Pp 236-238	CO3	Lab Submission and Class Test-2 and TEE
14	Multiplexers Types of MUX-2:1 MUX, 4:1 MUX, 8:1 MUX and 16:1 MUX		DEVELOP	TB1, TB2 TB1: Sec-4.11 Pp 158 TB2: Sec-6.1-6.2	CO3	-

	Designing of digital logic using multiplexers Multiplexers (4:1, 8:1 and 16:1)			Pp -231-237		
15	De Multiplexers (1:4, 1:8 and 1:16)- Truth table and circuit diagram		IMPLEMENT	TB2 TB2: Sec-6.3 Pp -238-241	CO3	Lab Submission and Class Test-2 and TEE
16	Encoders and Decoders- Truth table and circuit		APPLY	TB1, TB2 TB1: Sec-4.9 – 4.10 Pp 150-157 TB2: Sec-6.3 Pp -238-241	CO3	
Unit 4: Sequential Logic Circuits						
17	Flip-Flops-Concept, working, truth table, Excitation and timing diagram for the following Flip-Flops JK Flip-Flop SR Flip-Flop	PPT, Whiteboard	UNDERSTAND	TB1, TB2, RB1 TB1: Sec-4.9 – 4.10 Pp 150-157 TB2: Sec-7.1-7.10 Pp -238-241 RB1- Sec-8.4 Pp-319-326	CO4	Lab Submission and Class Test-2 and TEE
18	Master Slave JK Flip-Flop		UNDERSTAND	TB2, RB1 TB2: Sec-7.4.1 Pp -286-288 RB1- Sec-8.7 Pp-333-337	CO4	Lab Submission and Class Test-2 and TEE
19	Flip-Flops-Concept, working, truth table, Excitation and timing diagram for the following Flip-Flops D Flip-Flop T Flip-Flop		UNDERSTAND	TB1, TB2, TB1: Sec-5.4 Pp 195-202 TB2: Sec-7.5-7.6 Pp -288-289	**CO4**	Lab Submission and Class Test-2 and TEE
20	**Mid Term Test II**					
21	Flip-Flop conversion JK to T Flip-Flop JK to D Flip-Flop	PPT, Whiteboard	APPLY	TB2, RB1 TB2: Sec-7.8.1 Pp -293-294	**CO4**	Lab Submission and Class Test-2 and TEE

				RB1- Sec-8.8 Pp-339-342		
22	Flip-Flop conversion SR to T Flip-Flop SR to D Flip-Flop		SOLVE	TB1,RB1 TB2: Sec-7.8.1 Pp -293-294 RB1- Sec-8.8 Pp-339-342	**CO4**	Lab Submissions and TEE
23	Registers: Serial input - serial output; serial input-parallel output; Parallel In -Parallel Out, Serial In -Serial Out.		UNDERSTAND	TB1, TB2, TB1: Sec-6.1-6.2 Pp 255-266 TB2: Sec-7.10.2 Pp -300-301	**CO4**	
24	Registers: Serial input - serial output; serial input-parallel output; Parallel In -Parallel Out, Serial In -Serial Out.	PPT, Whiteboard	UNDERSTAND	TB1,TB2 TB1: Sec-6.1-6.2 Pp 255-266 TB2: Sec-7.10.2 Pp -300-301	**CO4**	-
25	Design of Asynchronous Counters-State Diagram, Truth table, Design and Circuit diagram		DESIGN	TB1 TB1: Sec-6.3 Pp 255-266	**CO4**	Lab Submissions and TEE
26	Design of Synchronous Counters- State Diagram, Truth table, Design and Circuit diagram		DESIGN	TB1,RB1 TB1: Sec-6.4 Pp 271-278	**CO4**	Lab Submissions and TEE

				RB1- Sec-10.10 PP 428-4339		
Unit 5: Introduction to VHDL						
27	VLSI Design flow: Design entry, Schematic, different modelling styles in VHDL: Dataflow, Behavioral and Structural Modelling	PPT, Whiteboard, TPS	DESIGN & IMPLEMENT	TB2 TB2: Sec-14.3.1-14.4.7 Pp 629-646	**CO5**	Lab Submissions and TEE
28	VLSI Design flow: Design entry, Schematic, different modelling styles in VHDL: Dataflow, Behavioral and Structural Modelling	PPT, Whiteboard	DESIGN & IMPLEMENT	TB2 TB2: Sec-14.3.1-14.4.7 Pp 629-646	**CO5**	Lab Submissions and TEE
29	Synthesis and Simulation of basic digital logic	PPT, Whiteboard, Flipped Classroom	DESIGN & IMPLEMENT	TB2 TB2:Sec-14.4.6-14.5.3 Pp 646 – 657 (Lec 9,10,11)	**CO5**	Lab Submissions and TEE
30	Synthesis and Simulation of basic digital logic		DESIGN & IMPLEMENT	TB2 TB2:Sec-14.4.6-14.5.3 Pp 646 – 657 (Lec 9,10,11)	**CO5**	Lab Submissions and TEE

Text Books

1. Morris Mano, Digital Design – With an Introduction to the Verilog HDL, VHDL, and System Verilog, 6th ed., Pearson Education, 2018.
2. R.P Jain, Modern Digital Electronics, 4th ed., Tata McGraw-Hill, 2013.

Reference Books

1. Kumar A. Anand, Fundamental of digital circuits, 4th ed., 2016.
2. D.P. Kothari and J.S. Dhillon, Digital Circuits and Design, Pearson Education, 2015.
3. Roth and John, Principles of Digital Systems Design, 6th ed., Cengage Learning, 2011.

4. Assessment and Evaluation Scheme

	Internal Continuous Assessment (ICA) (50 Marks)

	Class Tests (20 Marks)		Term-work (30 Marks)		
Assessment Component	Class Test-1	Class Test-2	Mini Project	Lab Performance	Lab Test
Marks	10	10	15	5	10

4.1 Internal Continuous Assessment (ICA) – 50 marks

Class Test-1 conducted in the 6th week. It will be for 10 marks (1-hour duration)..

Class Test-2 conducted in the 11th week. It will be for 10 marks (1-hour duration).

Lab Test: Student has to give lab test which will based on experiment 1 to experiment 10 performed in lab session

Lab Performance: Practical Lab output and time to time submission (One week) (05 Marks)

Mini Projects: project to be done in the lab as a group exercise. Mini Project Application Based Project (Team of 2-3) Hardware/Software Should address an application of Digital logic design.

4.2 Term End Examination (TEE) – (100 marks scaled down to 50)

TEE conducted at the end of the semester will be for 100 marks (3-hour duration). Marks obtained scaled down to 50. There will be 7 questions (each question will be 20 marks). Q1 will be compulsory and any 4 from the remaining to be solved.

4.3 Course Passing Criteria

4.3.1 ICA (50 marks) – No minimum marks

4.3.2 TEE (100 marks scaled to 50) – 40% required for passing

4.3.3 (ICA + TEE) (100 marks) - 40% required for passing

4.4 Assessments and Mapping to Course Outcomes

	Internal Continuous Assessment (ICA)					Term End Examination (TEE)
Course Outcomes	CT-1	CT-2	Mini Project	Lab Performance	Lab Test	TEE
CO-1	Y		Y	Y	Y	Y
CO-2	Y		Y	Y	Y	Y
CO-3		Y	Y	Y	Y	Y
CO-4		Y	Y	Y	Y	Y

CO-5		Y	Y	Y	Y	Y

5. Laboratory details

The following 10 laboratory exercises will form the submission for the laboratory coursework. Each exercise will contain 2 to 3 activities.

Sr. No.	Week No.	List of Lab Exercises	Mapped CO
1	1 & 2	a. To verify the behavior of all basic logic gates and universal gates using truth table and familiarization with digital integrated circuits. b. Realization of logic expression using basic logic gates and universal gates. c. Design a combinational logic circuit truth table for a given application.	CO1
2	3	To Verify NAND and NOR gate as Universal Gates. a. Implement NOT, AND, OR and X-OR using NAND gate b. Implement NOT, AND, OR and X-OR using NOR gate	CO2
3	4	a. Implement and verify De-Morgan's first theorem using logic gates b. Implement and verify De-Morgan's ssecond theorem using logic gates	CO2
4	5	a. Implement a binary to gray code converter, for a given binary number b. Implement a gray to binary code converter, for a given gray code	CO1 &CO3
5	6	To design and analyze adder circuits- a. Implement a half adder using minimum number of basic gates. b. Implement a full adder using minimum number of gates.	CO3
6	7	a. Study of 74XX… MSI digital multiplexers. b. Realize a given function using an appropriate MUX.	CO3
7	8	a. Implementation and verification of the following flip flops using logic gates and ICs: SR FF/ D FF/T FF b. Develop D FF using JK FF (Conversion of FF)	CO4
8	9	**Lab Test based on performed experiments and project**	
9	10	**Mini Project Discussion /Topic Finalization/Research Paper Idea discussion based on Project**.	
10	11	a. Design and implement an asynchronous binary counter using T FF b. Implement Counter using Counter IC.	CO4
11	12	. Write a VHDL Code to implement the following gates using Vivado	CO5

		a. AND Gate b. OR Gate c. NAND Gate d. X-OR Gate Write a VHDL Code to implement Binary to Gray Code convertor.	
12	13	Write a VHDL Code to implement Half and Full Adder.	CO5
13	14	**Mini project Test- Demonstration and Viva based on Project**	CO1-CO5
14	15	**Mini project Test- Demonstration and Viva based on Project Internal continuous assessments (ICA) evaluation and student submissions.**	-
15	16	Final Lab Submission with Index verification	-

6. Tutorial Plan

This course does not have any tutorial.

7. Course Material

References and Lab Manuals would be uploaded on LMS by faculty every week.

8. GenAI Usage

8.1 Pre-class Activity:

e.g. AI-Powered Boolean Expression Simplification:

- Activity:
  - Provide students with complex Boolean expressions or logic circuits that need to be simplified.
  - Have students use AI tools or online platforms capable of Boolean algebra simplification and Karnaugh map solving (such as an AI-enhanced Boolean expression simplifier).
  - Each student or group will simplify their assigned Boolean expressions both manually and using the AI tool.
  - Students will then compare the results from manual simplification with those generated by the AI tool, discussing any discrepancies and the efficiency of each method.

8.2 In-Class Activity:

e.g. Automated Logic Simplification:

- Activity: Use AI to simplify Boolean expressions or logic circuits.

- Objective: Help students understand the principles of logic minimization and optimization.
- Tools: Online AI tools for Boolean algebra simplification or Karnaugh map solvers.
- Discussion: Compare manual simplification processes with AI-based simplifications and discuss the implications for design efficiency.

Virtual Labs with AI Guidance:

- Activity: Conduct virtual labs where AI provides real-time guidance and feedback on digital logic design experiments.
- Objective: Provide a hands-on learning experience with immediate support from AI.
- Tools: Virtual lab platforms with AI integration, such as NI Multisim Live or Tinkercad.
- Discussion: Compare the virtual lab experience with traditional lab sessions and discuss the advantages and challenges

.Academic Integrity Statement

- Original work expected from students for all of the assigned assessment work. Copying in any form not acceptable and will invite strict disciplinary action. Evaluation of corresponding component will be affected proportionately in such cases. Plagiarism detection software will be used to check plagiarism wherever applicable. Academic integrity is expected from students in all components of course assessment.

Appendix

Assignment

Assignment Questions.docx